

Milestone 6: Frontend Development

Objective

The objective of Milestone 6 was to design and implement a complete **frontend dashboard** for the CreditPathAI system, integrate it with the trained machine learning backend, and validate the system through **User Acceptance Testing (UAT)**. This milestone ensures that the solution is usable by loan agents and decision-makers in a real-world scenario.

System Overview

The CreditPathAI application follows a **client–server architecture**:

- **Frontend:** React.js application for user interaction and visualization
- **Backend:** FastAPI service hosting the trained ML model
- **Visualization:** Plotly.js for real-time risk visualization
- **Model:** Logistic Regression–based credit default prediction model

The system enables agents to input loan application details and instantly receives:

- Default risk probability
- Risk categorization
- Actionable loan recommendations

Frontend Development

The frontend was developed using **React.js** and structured into reusable components to ensure scalability and maintainability.

Key UI Components

1. Loan Application Form

- Captures applicant and loan details:
 - Loan Amount
 - Interest Rate
 - Debt-to-Income (DTI) Ratio
 - Credit Score
 - Employment Duration (Months Employed)

2. Risk Visualization (Plotly Gauge Chart)

- Displays default probability as a percentage
- Dynamic gauge updates in real time after prediction
- Visual thresholds help agents quickly interpret risk severity

3. Agent Recommendation Panel

- Displays:
 - Risk Score (%)
 - Risk Level (Low, Moderate, High, Very High)
 - System-generated recommendation

Backend Integration

The React frontend communicates with the FastAPI backend using **RESTful API calls**.

Prediction Workflow

1. User submits loan details via the dashboard
2. Frontend sends a JSON request to the /predict endpoint

3. Backend:

- Applies data preprocessing
- Passes input through the trained ML model
- Computes default probability

4. Backend returns:

- Risk score
- Risk category
- Recommendation

5. Frontend updates the UI instantly without page reload

Agent Recommendation Logic

Based on the predicted default probability, the system categorizes risk and provides actionable recommendations:

Risk Level	Default Probability	Recommendation
Low Risk	< 25%	Approve loan
Moderate Risk	25% – 50%	Approve with monitoring
High Risk	50% – 75%	Approve with higher interest rate
Very High Risk	> 75%	Reject loan or demand collateral

This ensures alignment with real-world lending practices.

User Acceptance Testing (UAT)

UAT was conducted using multiple realistic borrower profiles representing different financial conditions.

Test Scenarios

- Low-risk borrower with high credit score and stable employment
- Moderate-risk borrower with average financial metrics
- High-risk borrower with high DTI and lower credit score
- Very high-risk borrower with poor credit history and short employment duration

UAT Outcomes

- The system consistently produced logical and explainable predictions
- Risk visualization updated accurately for all test cases
- Recommendations aligned with business and financial risk logic

Result:

UAT confirmed that the system functions correctly and meets stakeholder expectations.

Final Deliverables

- Fully functional **React.js frontend**
- Integrated **FastAPI backend**
- Real-time **Plotly.js risk visualization**
- Agent recommendation dashboard
- Successful UAT validation with documented results

Rollout Plan

1. Deploy backend API on a secure server
2. Host frontend on a web server or cloud platform
3. Provide basic training to loan agents on dashboard usage
4. Monitor system performance and collect feedback
5. Plan for future enhancements such as:
 - Model retraining
 - Additional financial features
 - Advanced analytics dashboards

Conclusion

Milestone 6 successfully delivered a complete, end-to-end **Credit Risk Assessment System**. The application demonstrates seamless integration between machine learning models and a user-friendly frontend, enabling loan agents to make informed, data-driven decisions. The system is production-ready and validated through comprehensive UAT, marking the successful completion of the project.